

Channel Knowledge Maps for FR3 UAV Base Stations: Dataset and ML-Based Modeling

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Abstract—This paper introduces a dense FR3 channel knowledge map (CKM) dataset for unmanned aerial vehicle (UAV) base stations. It contains 16180 ray-traced layout–height combinations from 3236 urban locations in 66 cities at 7.125 GHz. Every 513×513 realization stores building geometry, geometric visibility, attenuation, delay spread, angular spread, and a variable transmitter height. As a focused first benchmark, we propose an interpretable height-aware attenuation model. Its line-of-sight branch combines free-space loss with a coherent image-source two-ray term and compact height/range calibration. Its non-line-of-sight branch combines a COST231-shaped basis with local morphology and topology-specific linear calibration. Under a strict city holdout, the frozen model reaches 1.9289 dB overall root mean square error, with 1.7370 dB in LOS and 3.5363 dB in NLOS, on 2590 maps from 14 unseen cities. Its median CPU inference latency is 60.2 ms per map. The result provides a transparent attenuation baseline and a physical anchor for later distribution-aware refinement of all three channel quantities.

Index Terms—channel attenuation, channel knowledge map, delay spread, path loss, radio map, unmanned aerial vehicle.

I. INTRODUCTION

Unmanned aerial vehicles (UAVs) carrying base stations can support future 6G networks through mobility and an increased probability of line of sight (LOS). Their deployment nevertheless requires fast, location-dependent channel information for placement and resource management. Channel knowledge maps (CKMs) address this need by indexing channel information by transmitter or receiver location [1], [2].

Dense CKMs can be measured or simulated. Measurements are costly to collect at scale, whereas deterministic ray tracing (RT) provides spatially complete labels at substantial computational cost. Learned surrogates such as RadioUNet and PMNet therefore use convolutional encoder–decoders to predict dense maps from city geometry and transmitter information [3], [4]. Existing datasets, however, predominantly use fixed terrestrial transmitters and omit spatially varying building height.

The upper mid-band frequency range, FR3, is being studied as a coverage/capacity bridge for 6G [5]. It is attractive for UxNB deployment, but blocked links remain relevant even with elevated transmitters. Assessing this setting requires height-aware channel data across diverse urban environments. To our knowledge, no public CKM dataset currently combines FR3, UxNB-specific geometry, and dense three-dimensional building information.

a) Limitations of the existing CKM datasets: Table I compares the learning inputs distributed by each dataset rather than the provenance of its source geometry. Here,

3236 location groups paired with five transmitter heights produce 16180 realizations. “Building-height raster” means that every building pixel retains a height, rather than only a binary footprint; it does not imply that every height was surveyed directly. We combine global products providing both footprint and height [9], [10], avoiding the sparse height tags in OpenStreetMap [11].

A UxNB CKM should cover a typical cell diameter of 300–500 m with a spatial resolution capable of capturing relevant variations over approximately 20–80 wavelengths [12]. Since UxNB heights can approach 500 m, their channels differ substantially from fixed terrestrial links. UrbanRadio3D includes receiver heights up to 20 m [6], but does not provide the continuously varying elevated-transmitter geometry considered here. Accurate building height is particularly important because it controls visibility and rooftop blockage.

b) Limitations of the existing CKM modeling approaches: Dense attenuation is commonly treated as unconstrained image-to-image regression. This asks large networks to relearn stable propagation structure, often under fixed terrestrial geometry, and obscures whether they transfer to unseen cities and continuously varying UxNB heights.

c) Beyond state-of-the-art contributions: In this paper, we address these gaps. The main contributions are:

- 1) We introduce a 7.125 GHz CKM dataset generated via ray-tracing simulations in 3,236 city layouts across 66 diverse cities. This dataset relies on global-scale 3D building data [9], [10]. UxNB applications are enabled by using five transmitter heights per layout, resulting in 16180 CKM samples.
- 2) We propose a height-aware channel attenuation prediction framework that separates LOS and NLOS propagation using local city geometry. Free-space, coherent two-ray, and COST231/Hata-derived terms are calibrated through compact lookup tables and ridge regression, enabling zero-shot transfer to unseen cities.
- 3) We assess its validity under a strict city holdout through accuracy, component ablation, and computational comparisons.

Notation: Bold uppercase symbols denote rasters or tensors, bold lowercase symbols denote vectors, and $\odot$ is element-wise multiplication.

TABLE I
COMPARISON OF DENSE CKM AND RADIO-MAP DATASETS.

Property	This work	UrbanRadio3D [6]	RadioMapSeer [7]	CKMImageNet [8]
Variable elevated Tx height	yes	no	no	no
Layout groups or maps	3236 groups	701	701	40000
Raster size(s)	513×513	256×256	256×256	32×32 – 128×128
Horizontal resolution	1 m	1 m	1 m	2 m
Channel quantities	Att. (PL), LOS, DS, AS	PL, DoA, ToA	PL	Gain, AoA/AoD, delay
Building-height raster included	yes	yes	no	no
Per-sample topology labels	3 and 6 classes	no	no	no

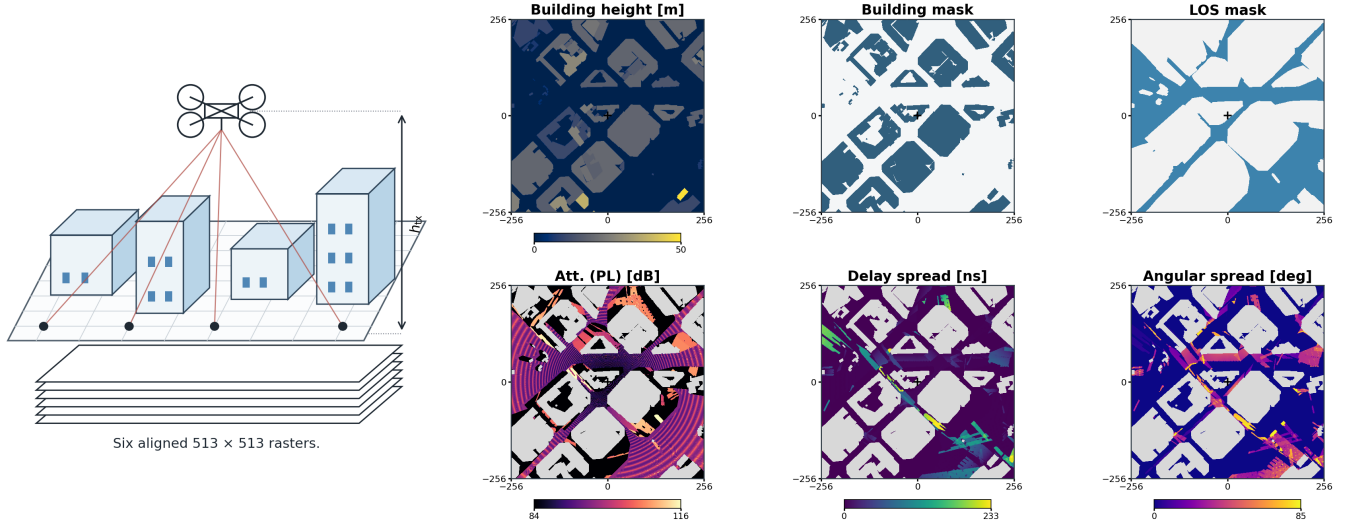


Fig. 1. One Barcelona realization (sample_02806) at $h_{tx} = 69.4$ m. The left illustration shows an elevated UAV base station serving ground receivers above six aligned 513×513 rasters. The panels show building height, building mask, LOS, attenuation, delay spread, and angular spread. Crosses mark the centered transmitter; gray areas are building pixels excluded from the metrics. For this sample, the stored three- and six-class labels are `dense_highrise` and `dense_block_midrise`.

II. CKM: DEFINITION AND DATASET DESCRIPTION

Figure 1 illustrates one sample from the dataset. In the following, UxNB and transmitter are used interchangeably.

A. CKM Definition and Mapping Formulation

For realization n , let $\mathbf{E}_n \in \mathbb{R}^{H \times W \times L}$ contain environmental channels, $\mathbf{C}_n \in \mathbb{R}^{H \times W \times F}$ contain channel quantities, and $\mathbf{m}_n \in \mathbb{R}^K$ contain metadata. CKM estimation seeks a mapping

$$\hat{\mathbf{C}}_n = \Phi_{\theta}(\mathbf{E}_n, \mathbf{m}_n), \quad (1)$$

where $H = W = 513$, $L = 2$, and $F = 4$.

B. The FR3 UxNB CKM Dataset

This subsection describes the environmental channels, ray-traced quantities, and metadata in each realization.

1) *3D Urban Environment Modeling*: The environmental tensor $\mathbf{E}_n$ combines GlobalBuildingAtlas and 3D-GloBFP [9], [10]. Unlike optional OpenStreetMap height tags, whose availability is below 10% [11], these products provide both building footprint and height. This information is necessary to represent how UxNB altitude changes rooftop blockage and visibility.

We extract city layouts of 3,236 urban locations in 66 diverse cities worldwide to cover a range of propagation topologies (from open low-rise suburbs to dense high-rise downtown areas). Each rasterized environmental map $\mathbf{E}_n$

spans $H = W = 513$ pixels at 1 m resolution. Its odd size centers the transmitter with symmetric offsets from -256 to 256 , while its wider field than 256×256 exposes more spatial propagation structure. The $L = 2$ environmental channels encode:

- Building height profile maps $\mathbf{H}$ indicating building height in meters (0 m over ground level).
- Binary building occupancy masks $\mathbf{B}$ indicating presence of structural obstacles.

2) *Ray Tracing Channel Simulation*: High precision ray tracing simulations are conducted to generate the spatially consistent channel tensors $\mathbf{C}_n$. To avoid using separate software for 3D model construction (e.g., Blender) and for RT (e.g., Wireless InSite or Sionna), SiteViewer and RT capabilities of MATLAB are used in this work.

RT simulation parameters are summarized in Table II. The 7.125 GHz frequency is a WRC-23 candidate band available across all ITU regions, while the 1 m resolution corresponds to approximately 24 wavelengths [12]. The UxNB is horizontally centered and evaluated at five independently sampled heights per location. The observed range is approximately 12 m to 478 m, yielding $N = 16180$ samples. Receivers are placed at 1.5 m over every non-building coordinate. The channel tensor stores $F = 4$ channels:

- Binary visibility mask $\mathbf{V}$: LOS (1) vs. NLOS (0) states.

TABLE II
RAY-TRACING SIMULATION PARAMETERS.

Parameter	Value
Carrier frequency, f	7.125 GHz
Antenna model	Isotropic
Raster size and resolution	513×513 pixels at 1-m resolution
Ground receiver height, h_{rx}	1.5 m
Observed transmitter height, h_{tx}	12 m to 478 m
UxNB heights per location	5
Total CKM samples, N	16,180

- Channel attenuation in dB.
- Multi-path root mean square (RMS) delay spread in ns.
- Multi-path RMS angular spread in degrees.

Building elements are assigned zero in $\mathbf{C}_n$ and are excluded from training and evaluation.

3) *Metadata*: Because carrier frequency, receiver height, and horizontal transmitter location (fixed at the grid center) are constant across all simulations, the metadata vector $\mathbf{m}_n$ consists of:

- Transmitter height h_{tx} , observed from approximately 12 m to 478 m.
- Categorical tags mapping the layout to three- and six-class morphology partitions. The attenuation model re-computes its three-class ITU route from the raster as described in Section III-B.

The three deployed ITU topology prototypes and the resulting support in the complete dataset are:

Topology	Samples	α	β [km^{-2}]	γ [m]
Suburban	7,435	0.1	750	8
Urban	7,350	0.3	500	15
Dense Urban	1,395	0.5	300	20

Here, “Samples” counts all 16,180 realizations. Each of the 3,236 layouts contributes its five transmitter heights to the same topology class.

The labeled HDF5 file stores both three- and six-class topology tags and the transmitter-height bin alongside the six rasters. Generation and model code are publicly available.¹

III. HEIGHT-AWARE CKM MODELING

In this section, we propose a propagation physics-informed CKM attenuation simulator (see Fig. 2). The model implements the attenuation-focused mapping function $\Phi_{\theta}(\mathbf{E}_n, \mathbf{m}_n)$ by modeling LOS and NLOS attenuation separately:

$$\mathbf{A} = \mathbf{V} \odot \mathbf{A}_{\text{LOS}} + (1 - \mathbf{V}) \odot \mathbf{A}_{\text{NLOS}}, \quad (2)$$

where $\odot$ denotes element-wise multiplication for masking. The LOS model uses free-space and coherent two-ray propagation corrected with terms learned from the CKM dataset. The NLOS model uses the ITU topology characteristics and a CKM-calibrated COST231-Hata basis. The visibility mask $\mathbf{V}$ is the binary output of the configured MATLAB ray tracer, not a prediction of the attenuation model; for matching

¹<https://github.com/unworthyzeus/CKMGenerator>; https://github.com/unworthyzeus/Final_Code_TFG.

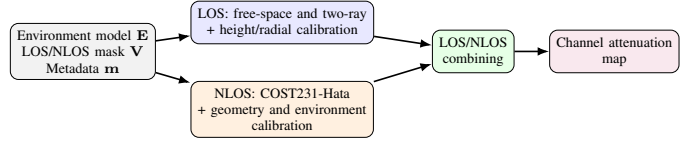


Fig. 2. The proposed solution uses separate LOS and NLOS propagation models calibrated with training CKM data.

geometry and RT settings, it can be stored once and reused. All calibration tables and regression weights are fitted using training cities and frozen before evaluating a held-out city.

Given the deployment geometry (h_{tx} , h_{rx} , and the fixed horizontal UxNB position), we pre-compute spatial geometry matrices over the 513×513 grid: horizontal distance $\mathbf{D}_{2D}$, direct 3D distance $\mathbf{D}_{3D}$, ground-reflected path length $\mathbf{D}_{GR}$, and elevation angle $\mathbf{\Theta}_{2D}$. For any receiver location $p = (i, j)$, the scalar geometric parameters ($d_{2D}, d_{3D}, d_{GR}, \theta$) are indexed directly from these matrices. For notational brevity, we omit the spatial index p in subsequent expressions.

A. Line-of-Sight Branch

For LOS propagation, the attenuation model combines direct free-space path loss Λ_{FSPL} with a coherent ground-reflection term $C_{2\text{ray}}$ to capture the two-ray propagation effects. Additionally, the model is calibrated based on CKM data as

$$\Lambda_{\text{LOS}} = \Lambda_{\text{FSPL}}(d_{3D}) + C_{2\text{ray}} + \zeta(h_{tx}) + r(d_{2D}, h_{tx}), \quad (3)$$

where $\zeta(h_{tx})$ is a height-dependent scalar bias correction and $r(d_{2D}, h_{tx})$ is a residual component capturing the effects introduced by horizontal distance and transmitter height.

The free-space term follows the Friis distance dependence, while the coherent correction retains the phase difference between the direct and reflected paths [13]:

$$\Lambda_{\text{FSPL}} = 20 \log_{10} \left(\frac{4\pi d_{3D} f}{c} \right), \quad (4)$$

$$C_{2\text{ray}} = -20 \log_{10} \left| 1 + \rho(h_{tx}) \frac{d_{3D}}{d_{GR}} e^{-j[k(d_{GR} - d_{3D}) + \phi(h_{tx})]} \right|, \quad (5)$$

where k is the free-space wavenumber, ρ is an effective reflected-field amplitude, and ϕ is a fitted phase offset. The calibration terms are obtained in two steps. Firstly, in each 5 m height bin, candidates $\hat{\rho} \in \{0, 0.05, \dots, 1.5\}$ and 48 uniformly spaced phases $\hat{\phi}$ in $[-\pi, \pi)$ are evaluated. For a candidate pair, the remaining bias is calculated as

$$\hat{\zeta}(\hat{\rho}, \hat{\phi}) = \frac{1}{|B_{\text{LOS}}|} \sum_{i \in B_{\text{LOS}}} \left[y_i - \Lambda_{\text{FSPL},i} - C_{2\text{ray},i}(\hat{\rho}, \hat{\phi}) \right], \quad (6)$$

where B_{LOS} is the set of valid LOS receivers and y is the target attenuation. The combination of $\hat{\zeta}, \hat{\rho}, \hat{\phi}$ minimizing the bin root mean square error (RMSE) is retained. Secondly, the remaining radial calibration term $r(d_{2D}, h_{tx})$ is calculated as the mean of the residual error.

In implementation terms, the LOS calibration is a lookup table that contains ρ, ϕ, ζ and r . Inference linearly interpolates neighboring height and radial entries.

B. Non-Line-of-Sight Branch

For shadowed receiver locations, our modeling approach relies on an empirical COST231-Hata baseline Λ_C , augmented by local structural features derived from the environment map **E**. The predicted NLOS attenuation is given by

$$\Lambda_{\text{NLOS}} = \mathbf{x}^\top \hat{\mathbf{w}}_q, \quad (7)$$

where $\mathbf{x} \in \mathbb{R}^{14}$ is a spatial context feature vector and $\hat{\mathbf{w}}_q$ is the height- and topology-dependent calibrated weight vector.

Evaluated at $f_{\text{MHz}} = 7125$, the COST231-Hata formulation is used as a starting point that requires calibration because its original domain does not cover the present frequency or transmitter heights [14]. Let $d_{\text{km}} = \max(d_{2\text{D}}/1000 \text{ m}, 0.001)$. The baseline is

$$\tilde{\Lambda}_C = 46.3 + 33.9 \log_{10} f_{\text{MHz}} - 13.82 \log_{10} h_{\text{tx}} - a(h_{\text{rx}}) + (44.9 - 6.55 \log_{10} h_{\text{tx}}) \log_{10} d_{\text{km}} + 3, \quad (8)$$

$$a(h_{\text{rx}}) = (1.1 \log_{10} f_{\text{MHz}} - 0.7) h_{\text{rx}} - 1.56 \log_{10} f_{\text{MHz}} + 0.8. \quad (9)$$

The regression uses the clipped baseline $\Lambda_C = \text{clip}(\tilde{\Lambda}_C, 0, 185)$, where $\text{clip}(x, a, b) = \min(\max(x, a), b)$.

To account for the local environment, we derive 13 contextual features from the building height grid **H** and building presence map **B** in **E**. Let $\mathbf{G} = 1 - \mathbf{B}$ be valid outdoor receiver locations and $\mathbf{N} = (1 - \mathbf{V}) \odot \mathbf{G}$ NLOS-ground locations. For $l \in \{15, 41\}$, let $\mathcal{W}_l(p)$ be the reflect-padded $l \times l$ window centered at p ; its box mean is $\mathcal{A}_l[z](p) = l^{-2} \sum_{q \in \mathcal{W}_l(p)} z(q)$.

The full feature vector $\mathbf{x}$ is constructed as

$$\mathbf{x} = [\Lambda_C, \ell_d, \delta_{15}, \delta_{41}, h_{15}, h_{41}, \delta_{41} \ell_d, n_{15}, n_{41}, n_{41} \ell_d, \sigma_{\text{sh}}, \theta', n_{41} \theta', 1]^\top. \quad (10)$$

where the components represent:

- Baseline and Distance: The COST231 baseline Λ_C and log-radial decay $\ell_d = \ln(1 + d_{2\text{D}}/1 \text{ m})$.
- Structural Context: Building density $\delta_l = \mathcal{A}_l[\mathbf{B}]$, normalized building heights $h_l = \mathcal{A}_l[\mathbf{H} \odot \mathbf{B}]/90 \text{ m}$, and shadowed ground fraction $n_l = \mathcal{A}_l[\mathbf{N}]$ evaluated within local ($l = 15$) and regional ($l = 41$) windows.
- Elevation Effects: Normalized path elevation angle $\theta' = \text{clip}(\theta/90, 0, 1)$ and the deterministic shadowing proxy $\sigma_{\text{sh}} = 2.3197[\max(90 - \theta, 0)]^{0.2361}$.
- Interaction Cross-Terms: Density–range ($\delta_{41} \ell_d$), blockage–range ($n_{41} \ell_d$), and blockage–elevation ($n_{41} \theta'$) interaction terms, alongside the scalar bias unit (1).

The topology route is computed from each height raster using the standard ITU parameters [15], [16]. Let $\mathcal{B}$ be the building pixels, let their four-connected components represent N_b footprints, let $A = 513^2 \times 10^{-6} \text{ km}^2$, and let H_j be the maximum raster height in footprint j . We estimate

$$\hat{\alpha} = \frac{|\mathcal{B}|}{513^2}, \quad \hat{\beta} = \frac{N_b}{A}, \quad \hat{\gamma} = \sqrt{\frac{1}{2N_b} \sum_{j=1}^{N_b} H_j^2}. \quad (11)$$

Thus, $\hat{\alpha}$ is the built-up fraction, $\hat{\beta}$ is the footprint density in buildings per km^2 , and $\hat{\gamma}$ is the fitted Rayleigh mode of

the footprint heights. For prototype k , the selector computes $d_k = \|\log[(\hat{\alpha}, \hat{\beta}, \hat{\gamma})/(\alpha_k, \beta_k, \gamma_k)]\|_2$ and chooses the smallest. Equal-weight log-ratios compare relative deviations and prevent the scale of β from dominating. The three deployed prototypes and their support are listed with the dataset meta-data. The urban high-rise prototype (0.5, 300, 50) is merged into dense urban because only 90 training maps select it.

The regime weight vectors $\hat{\mathbf{w}}_q$ are calibrated offline using CKM training samples in two steps. Firstly, scenes are partitioned into nine regimes from the combinations of three ITU topologies and three standard transmitter altitude groups: low ($h_{\text{tx}} \leq 60 \text{ m}$), medium ($60 \text{ m} < h_{\text{tx}} \leq 100 \text{ m}$), and high ($h_{\text{tx}} > 100 \text{ m}$). Secondly, for each regime q , at most 1024 NLOS channel realizations per training sample are selected deterministically and fed to the ridge model. For training stability during offline calibration, let $\mathbf{z}_i$ denote the standardized counterpart of $\mathbf{x}_i$:

$$\hat{\mathbf{w}}_q = \arg \min_{\mathbf{w}} \sum_{i \in q} (\mathbf{z}_i^\top \mathbf{w} - y_i)^2 + 10^{-2} \|\mathbf{w}_{-b}\|_2^2. \quad (12)$$

Here, $\mathbf{z}_i$ contains standardized non-intercept features, and the L_2 term penalizes only $\mathbf{w}_{-b}$ while omitting the bias (intercept). After transforming the coefficients back to raw feature space, the deployed estimate is clipped to the stored attenuation support:

$$\Lambda_{\text{NLOS}} = \text{clip}(\mathbf{x}^\top \hat{\mathbf{w}}_q, 20, 185). \quad (13)$$

IV. EXPERIMENT

A. Experimental Setup and Evaluation Protocol

The dataset split contains 10,840 maps from 37 training cities, 2,750 maps from 15 validation cities, and 2,590 maps from 14 final test cities. No city is moved between these partitions to ensure strict spatial generalization. The calibration procedure described in Section III uses only training cities, where all nine topology and height regimes are fitted on the 10,840 CKM training samples. The validation subset is reserved for development checks and component ablation. Final performance results use exclusively the unseen 14-city test subset.

After excluding building interiors and outdoor channels lacking valid ray tracing targets, the test dataset contains 423 087 119 valid ground receivers: 391 721 805 LOS and 31 365 314 NLOS. Thus, 7.413% of the evaluated receivers are NLOS because of the elevated UxNB geometry. Although the pixel-weighted overall score is therefore LOS-dominated, 424 of the 2,590 test maps contain more than 25% valid NLOS receivers, motivating separate regime-specific evaluation. An example of the model output is shown in Fig. 3.

B. Prediction Accuracy and Height Generalization

The prediction performance on the unseen test set across the standard ITU UxNB altitude regimes is summarized in Table III. Overall, the proposed height-aware simulator achieves a global RMSE of 1.9289 dB (1.7370 dB for LOS and 3.5363 dB for NLOS). Per-city overall RMSE ranges from 1.6149 dB in Halifax to 2.5851 dB in Osaka, with

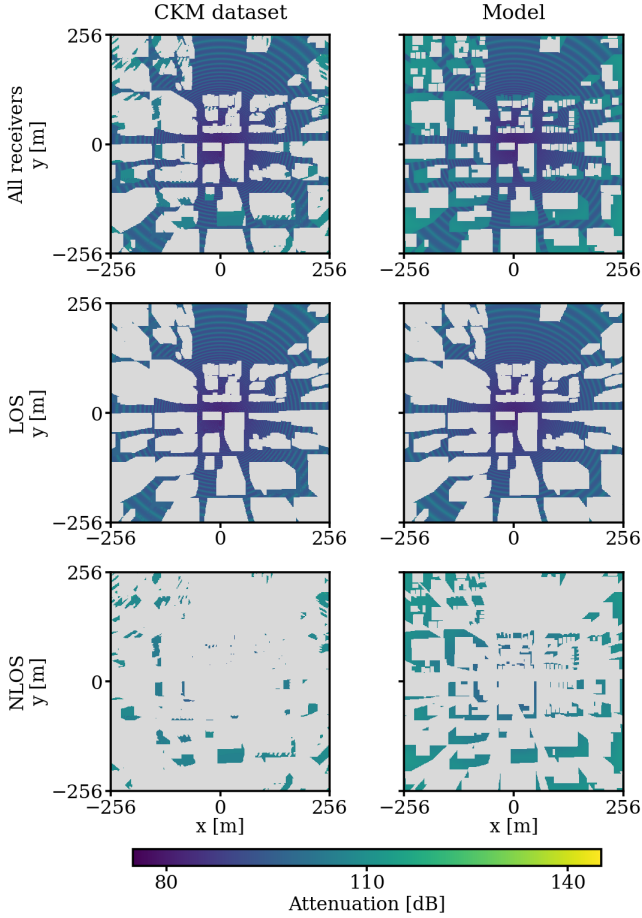


Fig. 3. Held-out Vancouver sample_15262 ($h_{tx} = 50.30$ m; urban, low-transmitter regime). The dataset column shows only stored valid targets, while the model column predicts every ground receiver. Thus, gray in the first dataset panel denotes buildings or an unavailable target; in the first model panel it denotes buildings only. Lower rows additionally exclude the other visibility class. Metrics use valid targets only. The map has 119180 LOS and 25607 NLOS targets; full, LOS, and NLOS RMSE are 2.18, 2.10, and 2.52 dB.

an unweighted 14-city mean of 1.9831 dB. Across the low, medium, and high altitude groups, overall RMSE decreases from 2.1437 dB to 1.7117 dB.

Furthermore, Table IV breaks down final test error by standard ITU topology. Suburban layouts exhibit the lowest overall error (1.6977 dB), while dense urban layouts present the harshest NLOS environment (4.1319 dB), validating topology-specific regime calibration. The fitted coefficient vectors also change substantially across topologies, so one global set of slopes would mix distinct relationships.

The GMM U-Net is a controlled neural alternative: a height-conditioned mixture head and U-Net decoder, separated into six topology classes and LOS/NLOS experts, are trained with the same training split and evaluated on the same 2590 test maps and masks as the proposed method. It was the best purely neural approach in an internal exploration of over 50 configurations. Its lower NLOS error shows where distribution-aware learning can help, while the lower LOS and overall errors of

TABLE III
RMSE [dB] BY STANDARD ITU UXNB ALTITUDE REGIME.

UxNB height	Overall	LOS	NLOS	Test maps
Low ($h_{tx} \leq 60$ m)	2.1437	1.7923	3.7526	899
Medium ($60 \text{ m} < h_{tx} \leq 100$ m)	1.9682	1.7747	3.5080	773
High ($h_{tx} > 100$ m)	1.7117	1.6697	2.7620	918
All heights	1.9289	1.7370	3.5363	2590

TABLE IV
RMSE [dB] BY STANDARD ITU TOPOLOGY.

Topology	Overall	LOS	NLOS	Test maps
Suburban	1.6977	1.6048	3.1632	1455
Urban	2.2963	1.9967	3.6466	940
Dense urban	2.5716	2.1126	4.1319	195
All topologies	1.9289	1.7370	3.5363	2590

the proposed model show the value of resolving the stable coherent component explicitly. A following paper will study a hybrid model using calibrated attenuation, delay spread, and angular spread priors as inputs and base estimates for neural refinement.

The remaining rows are published context, not a controlled ranking. RadioUNet and RadioVAE [3], [17] use 256×256 fixed-height terrestrial RadioMapSeer data; TSRDiff [18] uses sparse targets, and RO-HMTGP [19] predicts measured RSRP rather than dense simulated attenuation. Their values indicate scale under different tasks, not superiority over the present 513×513 variable-height city holdout.

C. Ablation Study

On frozen test data, coherent reflection lowers LOS RMSE from 3.9925 dB for raw free-space loss to 1.7424 dB; fitted coefficients and topology/height calibration lower raw COST 231 NLOS RMSE from 25.0098 dB to 3.5363 dB. A trial-and-error ablation, with every candidate recalibrated, found that an eight-coefficient model using the 41-pixel morphology scale preserves over 90% of the complete model's improvement over one global NLOS constant on test; intermediate results are omitted.

D. Computational Complexity

Table VI compares the proposed simulator with MATLAB ray tracing and neural baselines whose published dimensions permit reproducible multiply-accumulate operation (MAC) counts. CPU timings use the same AMD Ryzen 5 5600X with GPU disabled; neural timings exclude file I/O and use 10 warmups and 50 forward passes. Given the visibility mask, the prior's median latency is 0.0602 s per 513×513 map, excluding file I/O.

The proposed state is 66800 LOS lookup values plus 126 NLOS coefficients; its time includes feature construction and LOS evaluation. MATLAB RT also returns visibility and three channel quantities, so the approximately $1700 \times$ gap is significant but not direct. Timings are orientative: a PyTorch

TABLE V
REPORTED RMSE [dB]; ONLY THE FIRST TWO ROWS SHARE A TEST CONTRACT.

Method	Evaluation contract	All	LOS	NLOS
Proposed	CKM city holdout	1.9289	1.7370	3.5363
GMM U-Net	Same split and masks	4.3606	4.4323	3.3381
RadioUNet	RadioMapSeer DPM	1.60	–	–
RadioVAE _C	RadioMapSeer DPM	1.42	–	–
TSRDiff	BRAT-Lab, 10% samples	3.62	–	–
RO-HMTGP	Measured RSRP	4.19–6.00	–	–

TABLE VI
COMPUTATIONAL CONTEXT ON THE SAME CPU.

Method	Grid	State	MACs	CPU [s]
Proposed	513 × 513	66.8k+126	0.170M*	0.0602
RadioUNet [3]	256 × 256	13.27M	12.73G	0.1339
PMNet [4]	256 × 256	33.34M	52.71G	0.4042
SSL-Radio [20]	256 × 256	15.4–20.4M	16.9–21.9G†	–
MATLAB RT	513 × 513	–	–	102.289

*Counts only the final NLOS dot products. †Reconstructed from the published layer table.

GPU port preserves the MAC reduction, but exact latency ratios require same-GPU tests.

E. Scope and Limitations

The reported accuracy is conditional on the supplied RT visibility mask. For fixed city geometry, transmitter position and height, receiver grid, and RT settings, that mask is deterministic and can be stored once; all 742392 values matched in a five-condition audit. The fitted tables inherit the simulator, antenna, carrier, terrain, and building products, while COST231 is only a calibrated basis outside its original domain. The result is therefore a transparent dataset baseline and refinement anchor, not a universal propagation law.

V. CONCLUSIONS

This paper introduced a dense FR3 (7.125 GHz) UAV CKM dataset providing attenuation, delay spread, angular spread, geometric visibility, building geometry, transmitter height, and topology labels across 16,180 simulated samples from 66 global cities. Using this dataset, we proposed a computationally efficient, height-aware attenuation prediction framework. By relying on conventional propagation models calibrated on the dataset, the model generates accurate attenuation maps and generalizes to unseen city layouts. It achieves an overall test RMSE of 1.9289 dB and executes in 0.0602 s per map on a CPU. The results further show that topology is the principal NLOS calibration partition, with transmitter height refining that partition and the regional morphology scale carrying most of the useful local context.

The model is intended as a physical/statistical base rather than a rejection of deep learning. The controlled GMM U-Net result identifies NLOS as the promising refinement target, but also shows the cost of asking a network to relearn the coherent LOS field. A future journal paper will train distribution-aware

networks to refine analogous calibrated priors for attenuation, delay spread, and angular spread. Its multi-output runtime will not be directly comparable to this attenuation-only benchmark, but the present computational gap shows that the prior-first strategy is significant and worth preserving.

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